

Transitioning to BWA networks

Transitioning to next-generation broadband wireless access networks requires upgrades in terms of the core network infrastructure, backhaul networks, devices and applications. A smooth transition also requires a clear migration path for operators.

1. **Radio access network.** UMTS, LTE and WiMAX require new RANs to be installed.
2. **Backhaul networks.** First to support the broadband access, we need to upgrade the speed of backhaul networks. From the current 10G Ethernet speed, it needs to be upgraded to 40G or even 100G. Upgrading to the highest speed will ensure that more base stations can be connected to the backhaul.
3. **Devices.** The availability of BWA devices at affordable costs will increase more penetration and acceptance of the technology. Users would typically require access devices such as USB dongles, mobile devices with built-in 4G modems (e.g., phones and iPad), or fixed wireless modems at home for BWA. Several products using WiMAX 802.16e are already available today, and 4G WiMAX products will also be available soon. In the case of LTE too, market leaders like Samsung, LG and Ericsson have launched a few chipsets for the pre-4G version, while a lot more chipsets and devices are currently in the development/trial stage for LTE-Advanced. The availability of LTE devices is on the rise, and prices are expected to see a downward movement from 2012 going forward.
4. **Applications.** The availability of rich applications that will completely utilise the abilities of BWA networks will speed up the adoption of such technologies.
5. **Migration path for 3G operators.** A clear migration path for 3G operators to move into 4G by reusing as much of the core and service infrastructure network elements needs to be clearly laid down. There is a clear migration path available for operators to move from 2G to 3G UMTS to LTE if they have adopted 3GPP family of products. Transitioning from 3G to 4G will be easier for operators using 3GPP technologies than WiMAX ones. WiMAX is more likely to be adopted by new operators than existing ones.

standards. Some advanced countries, however, are already in the maturity stage of UMTS/HSPA standards and in the initial stages of 4G WiMAX 802.16m and 3GPP LTE.

The strongest trend in the future is clearly for the LTE standard. Apart from using the latest techniques and technologies in modulation, multiple access, coding, routing, radio and antennas, LTE has the additional advantage of being the only preferred 4G cellular standard globally with support of all major operators, and has possession of the lucrative lower-level spectrum (e.g., 700 MHz in many countries). All of this makes it a very strong force in the BWA revolution.

Dr Borkar confirms, "In the foreseeable future the global market seems to be almost equally divided between GSM/GPRS/EDGE, WCDMA/HSPA, LTE, and WiMAX with the first two declining, LTE expanding, and WiMAX comparatively stable or marginally increasing."

"Whatever standard wins, there

will be co-existence of UMTS/HSPA/HSPA+, WiMAX, Wi-Fi and LTE, and seamless session mobility and interoperability (at packet core) to meet the user-demand will be the way forward. Wireless spectrum availability always is a challenge and to address this, leveraging on all the access technologies will be the need of the hour," says Srinivasan.

Does India have a special need for BWA?

Would India benefit in any specific way by the adoption of BWA technologies? Of course, say the experts. BWA will be very beneficial in covering the interior regions of India that are not yet covered by wired lines.

"Not only the interiors, but entire India has the need for BWA since even in major cities, wired network has very limited coverage. As part of leap-frogging technology from the current outdated systems (34 million wire-line customers across the country and the wireless technology being the 2G technology from the early 1990s),

BWA is imperative for India if the country wants to grow. Depending upon the strategies of investment and migrations, the 3GPP WCDMA/HSPA is very appropriate at the current time followed by the 3GPP LTE at a suitable time in future, augmented by WiMAX 802.16m at selected places. The key issue for deployment is the backhaul. In addition to the access and the core infrastructure, backhaul is needed across all entities—access to core as well as across the whole public data network (PDN). Setting up an effective PDN is another major challenge," says Dr Borkar.

He adds that in India, the transition to next-generation BWA networks is likely to be driven by service providers, who in turn are driven by consumer demand for data services and average revenue per user (ARPU) from data. The recent trends show that the demand for mobile data is showing a steady growth in the country. Therefore it is expected that the latest 4G technologies will be introduced in India in two to three years in the face of growing consumer demand. The challenges for service providers are more likely to be spectrum availability and short-term incentives to invest, particularly in rural areas where the ARPU is likely to be lower than in urban areas.

Shyam Ananthnarayan, vice president-communications and wireless technology business, Tata Elxsi, sums up, "BWA promises to help every section of the society through better access to information, education and new employment opportunities to Indians in both urban and rural areas. India has a special need for BWA to connect the people of every village, town, semi-rural, semi-urban and urban area to bring them on to one platform. It is expected that both WiMAX and LTE will find a place in a wireless nation. While LTE seems to be a natural migration path for operators in urban areas, WiMAX is likely to find a place in semi-urban and rural areas of the country." ●

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